INCH-POUND

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22 June 2005

COMMERCIAL ITEM DESCRIPTION

GEAR, FIRE PROTECTIVE

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description covers the requirements for a structural fireman's coverall to be used by Navy damage control fire fighters.

2. CLASSIFICATION. The Fire Protective Gear (FPG) shall conform to the following sizes.

2.1 Sizes. Eighteen (18) sizes are available: Small-Short, Small-Regular, Small-Long, Medium-Short, Medium-Regular, Medium-Long, Large-Short, Large-Regular, Large-Long, XLarge-Short, XLarge-Regular, XLarge-Long, XXLarge-Short, XXLarge-Regular, XXLarge-Long, XXXLarge-Short, XXXLarge-Regular, XXXLarge-Long. (see Table I, 2.1.1, and 2.2.2)

TABLE I. FPG Sizes and corresponding measurements (in inches minimum).

SIZE	BACK RISE	CHEST	FRONT RISE	HIP	REGULAR SLEEVE LENGTH	THIGH	WAIST
Small	37 ½	45	31 ½	47	33	28 ¾	31 ½
Medium	38 ½	49	32 ½	51	34 ½	30 ¾	35 ½
Large	39 ½	53	33 ½	55	35	32 ¾	39 ½
XLarge	40 ½	57	34 ½	59	35 ½	34 ¾	43 ½
XXLarge	41 ½	61	35 ½	63	36	36 ¾	47 ½
XXXLarge	42 ½	65	36 ½	67	36 ½	38 ¾	51 ½

Notes:

1. Leg zipper opening for all sizes is 18 inches minimum.
2. All measurements are taken with shell only.
3. Back rise is measured from crotch to collar line.
4. Chest is measured across the garment at the underarm and measurement is doubled.
5. Front rise is measured from crotch to collar line.
6. Hip is measured across the garment 8 inches below the waist and the measurement is doubled.
7. Sleeve length is measured from center back to bottom of cuff.
8. Thigh measurement is taken at crotch height across a leg and measurement is doubled.
9. Waist measurement is taken at the waistband stretched out and measurement is doubled.

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FSC 4210

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2.1.1 Inseams. Inseams shall be of the following lengths:

a. Short – 27"

b. Regular – 29"

c. Long – 31"

2.1.2 Sleeve length. Sleeves shall be of the following lengths: Short sleeves shall be 2" less than regular and long sleeves shall be 2" greater than the regular for the sizes listed in Table I above.

3. SALIENT CHARACTERISTICS.

3.1 Design and construction. The fire protective gear (FPG) (see Figure 1) shall be a one-piece garment with a snap-in removable liner system. The liner system shall be composed of a moisture barrier (liquid/fluid) and a thermal barrier. The FPG shall be a raglan sleeve design with side body gussets. The FPG shall have a front zipper closure, elastic waistband, two leg zippers, sleeve cuffs, leg cuffs, two cargo pockets with flap closures 8 x 7.5 x 2 inches; 2-inch reflective trim; reinforcements for the knee, elbow, leg cuff and cargo pocket; over-the-thumb wristlets; and a stand-up collar. Each wristlet shall be attached to the liner system, be a minimum length of 8 inches, and shall contain a thumbhole with a minimum diameter of 1 inch. The sleeve cuff of the outer shell shall be a partially gathered elastic cuff with a hook and pile tab closure. The elbow and bottom half of the cargo pockets shall be reinforced. The shoulder area shall be four inches minimum from the center of the shoulder towards the front and back of the garment, and shall provide additional heat protection to meet the requirements of 3.2.2. The reinforced knee area shall be at least a 9-inch by 9-inch area and shall provide additional heat protection to meet the requirements of 3.2.2. The FPG and all layers of the FPG shall be individually graded per size. The two cargo pockets shall be located one on each leg above the knee area.

3.1.1 Color. The color of the outer shell shall be natural. The knee/leg cuff reinforcements shall be gray in shade. The color of the reflective trim shall be yellow.

3.1.2 Yarns. All fabrics and battings used in the construction of the outer shell, moisture barrier, and thermal barrier shall be manufactured from inherently flame-resistant yarns. All other materials used in the construction of the garment, such as zipper tapes, collar linings, elastic, hook and loop tapes, padding, reinforcements, interfacing, bindings, hanger loops, emblems, labels, patches, and trim, shall meet the requirements of NFPA 1971.

3.1.3 National Fire Protection Association (NFPA). The FPG shall be compliant with NFPA 1971 and shall be third party certified to this standard.

3.2 Performance requirements.

3.2.1 Thermal. The FPG shall have a minimum Thermal Protective Performance (TPP) rating of 40 when tested in accordance with NFPA 1971.

3.2.2 Conductive/compressive heat resistance (CCHR). The knee and shoulder shall have a minimum heat transfer index of 100 for the knee and 65 for the shoulder areas of the FPG. The testing shall be in conformance with the test procedure as follows:

a. Samples shall measure 152.4 mm x 152.4 mm ±6.4 mm (6 in. x 6 in., ±¼ in.).

b. Samples for the knee and shoulder areas shall be representative of the composites used in those areas of the FPG.

c. A total of six specimens shall be tested. Three specimens from the knee area and three specimens from the shoulder area shall be tested.

d. Specimens shall be preconditioned in accordance with standard textile laboratory conditions specified in ASTM D1776.

e. Sample specimens shall be tested in accordance with ASTM F1060 with the modifications specified herein.

f. Specimens shall be tested using an exposure temperature of 280 ºC (536 ºF).

g. The pressure applied during the test shall be 0.008 kg/cm^2 ±0.0008 kg/cm^2 (2 psi, ±0.2 psi) for shoulder and 0.032 kg/cm^2, ±0.0032 kg/cm^2 (8 psi ±0.8 psi) for knee areas.

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h. The CCHR rating shall be the time in seconds to achieve a temperature rise of 24 ºC ±2 ºC (43 ºF ±3.6 ºF) using a starting temperature of 20.0 ºC ±5 ºC (68 ºF ±9 ºF) for all specimens. The time shall be measured to the nearest tenth of a second.

i. The individual CCHR rating for each specimen shall be reported. The average CCHR rating shall be calculated and reported.

j. If an individual result from any test set varies more than ±8 percent from the average results, the results from the test set shall be discarded and another set of specimens shall be tested.

3.2.3 Outer shell.

3.2.3.1 Weight. The outer shell shall have a maximum weight of 7.9 ounces per square yard when tested in accordance with ASTM D3776, Option C.

3.2.3.2 Strength. The outer shell shall have a 22-pound retained strength after heat exposure. The testing shall conform to the following test procedure:

a. A sample of the outer shell by itself shall be placed on the thermal performance tester in accordance with NFPA 1971.

b. The test shall be performed with the following exceptions:

(1) Only standard laboratory preconditioning shall be performed in accordance with ASTM D1776.

(2) A 17.5-second exposure shall be performed.

(3) The recording of results shall not be accomplished.

(4) The testing shall be conducted on five specimens.

(5) The only weight on the sample shall be the sensor block; no additional weight shall be used.

c. Following the 17.5-second exposure, the exposed samples shall be tested for burst strength in accordance with ASTM D6797.

d. The average shall not be lower than the requirement as stated above.

3.2.3.3 Weight loss. The outer shell fabric shall have a maximum weight loss after abrasion of 4.0%. The testing shall be in accordance with ASTM D3884 and shall be performed with the following exceptions:

a. All samples shall be weighed to the one hundredth place prior to abrasion.

b. Samples shall be run for 500 cycles with the H18 wheels and adding auxiliary weights, 500 grams per wheel, for a total weight added by two wheels of 1000 grams applied to the specimen.

c. Taber abrasion machine shall be run with vacuum in off position. Loose particles shall be air blown cleaned from each sample after testing, and then re-weighed and percent weight loss calculated.

d. Cleaning of each wheel shall take place after each individual specimen is run 500 cycles. Periodic cleaning of wheels during the 500 cycles may be conducted as needed to maintain a relatively clean wheel surface. Cleaning shall be conducted with a small stiff brush, applying minimal to no pressure to the running wheel.

e. Upon completion of each fabric sample set (N=5), each H18 surface grit wheel shall be resurfaced in accordance with ASTM D3884. Utilize resurface disc material of 150-grit. Resurfacing duration shall be 25 cycles. After resurfacing, clean the abrasion wheels with a vacuum to remove dust. The resurfacing disc should be replaced after 50 total cycles (good for 2 sets total).

3.2.4 Moisture barrier. The moisture barrier shall have a water vapor diffusion resistance no greater than 650 seconds per meter (S/M) when tested in accordance with ASTM F2298, Method A. The specimens shall be tested in the cell for a minimum of one hour.

3.2.5 Thermal barrier. The liner face cloth shall have a minimum of 33% by weight filament yarns (certificate of compliance). The weight of the total barrier system shall have a maximum weight of 8.3 oz/sq yd when tested in accordance with ASTM D3776.

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3.2.6 Reflective trim. The reflective trim shall be two inches in width and there shall be one band around the sleeve, trouser leg, chest, and below the waistband. The coefficient of retro reflection after heat exposure shall be a minimumvalue of 100 cd/lux/m2 when tested as follows:

a. The specimens shall be preconditioned by performing the heat and thermal shrinkage resistance test in accordance with NFPA 1971. Thermal shrinkage shall not be measured.

b. The specimens shall be tested for retro reflection after being preconditioned in accordance with NFPA 1971.

3.2.7 Outer knee/leg cuff reinforcement material. The material shall contain a high solids polymer face coating for protection against abrasion. The maximum weight shall be 20.0 oz/sq yd when tested in accordance with ASTM D3776. The maximum thermal shrinkage in either direction shall be 1.0% when tested in accordance with NFPA 1971. The maximum vertical flammability shall be no more than 1 second after flame and a maximum char length of 1.0 inch when tested in accordance with ASTM D6413.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 ASTM. ASTM standards are available from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or online at www.astm.org/.

7.1.2 FAR. The Federal Acquisition Regulation (FAR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at http://www.arnet.gov/far.

7.1.3 NFPA. National Fire Protection Association (NFPA) standards are available from NFPA, 1 Batterymarch Park, Quincy, MA 02169-7471 or online at www.nfpa.org.

7.2 Ordering data. The contract or order should specify the following:

a. CID document number and revision.

b. Product conformance provisions.

c. Quantity required.

d. NSN or size ordered.

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7.3 National stock numbers (NSNs). The following is a list of NSNs assigned that correspond to this CID. The list may not be indicative of all possible NSNs associated with the CID.

NSN	ITEM	NSN	ITEM
4210-01-468-5505	Small/Short FPG	4210-01-468-5599	XLarge/Short FPG
4210-01-468-5528	Small/Regular FPG	4210-01-468-5671	XLarge/Regular FPG
4210-01-468-5539	Small/Long FPG	4210-01-468-5673	XLarge/Long FPG
4210-01-468-5545	Medium/Short FPG	4210-01-468-5677	XXLarge/Short FPG
4210-01-468-5551	Medium/Regular FPG	4210-01-468-5679	XXLarge/Regular FPG
4210-01-468-5552	Medium/Long FPG	4210-01-468-5681	XXLarge/Long FPG
4210-01-468-5562	Large/Short FPG	4210-01-468-5683	XXXLarge/Short FPG
4210-01-468-5565	Large/Regular FPG	4210-01-468-5689	XXXLarge/Regular FPG
4210-01-468-5568	Large/Long FPG	4210-01-468-5691	XXXLarge/Long FPG

7.4 Key words.

Firefighting coverall
Firefighting ensemble
Firefighting suit

MILITARY INTERESTS

Custodians:
Navy – SH
Air Force – 03

Preparing Activity:
Navy – SH
(Project 4210-0662-000)

Review Activities:
Navy – AS
Air Force – 84, 99
DLA – IS

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FIGURE 1. Fire protective gear.

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